

s_lo1rECB: ECB Intervention and Simulation Results

May 26, 2026

ECB Model: Sovereign Problem

$$V^{ND}(b, g, s, \varepsilon) = \max_{b'} \left\{ u(c^{ND}) + \beta g^{1-\gamma} \mathbb{E} \left[\max \{ V^{ND}(b', g', s', \varepsilon'), V^D(b', g', s', \varepsilon') \} \mid g, s \right] \right\}$$

$$c^{ND} = y(g, \varepsilon) - b + n(b', g, s), \quad y(g, \varepsilon) = g e^{\sigma_\varepsilon \varepsilon}, \quad n(b', g, s) = \frac{g b'}{R(b', g, s)}$$

$$V^D(b, g, s, \varepsilon) = u(\phi_g y(g, \varepsilon)) + \beta g^{1-\gamma} \mathbb{E} \left[\theta V^D \left(\frac{b}{g}, g', s', \varepsilon' \right) + (1 - \theta) \max \left\{ V^{ND} \left(\kappa \frac{b}{g}, g', s', \varepsilon' \right), V^D \left(\frac{b}{g}, g', s', \varepsilon' \right) \right\} \mid g, s \right].$$

Key Difference From Direct LOLR Lending

There is no sovereign debt state ℓ : the ECB buys private bonds after market clearing rather than lending directly to the sovereign.

ECB Facility and the Private-Bond Schedule

$$Q(b, g, s, \varepsilon) = (1 - d) + dX, \quad H(b', g, s) = \frac{\mathbb{E}[Q(b', g', s', \varepsilon') \mid g, s]}{1 + r^*}$$

$$P(g, s) = \begin{cases} \frac{1}{1 + r^* + \sigma^\ell}, & \text{repayment, } g = g_L, s = s_b, \\ 0, & \text{otherwise,} \end{cases} \quad L(b', g, s) = \pi P(g, s) + (1 - \pi)H(b', g, s)$$

$$R(b', g, s) = \frac{1}{\max\{H(b', g, s), L(b', g, s)\}}, \quad n(b', g, s) = \frac{gb'}{R(b', g, s)}$$

Intervention

Only lenders holding bonds issued in repayment, low growth, and the bad sunspot may use the facility. They queue only if $L > H$.

Reported Policy Grid

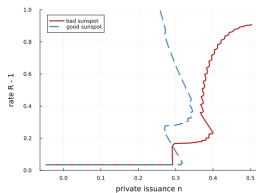
$$\pi \in \{0, .25, .50, .75, 1\},$$

$$\sigma^\ell \in \{.04, .06, .08, .10\}.$$

Baseline: $\pi = .50, \sigma^\ell = .04$.

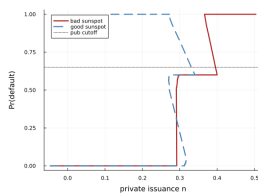
Baseline Policy: Private Schedule and Issuance

ECB model: accessible private schedule, low growth

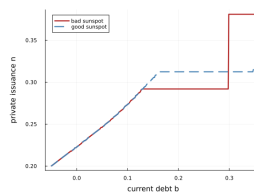


Selected private schedule in low growth

Default probability on selected schedule

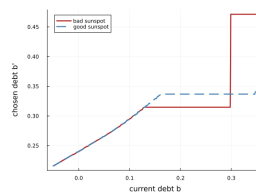


Private issuance policy, low growth



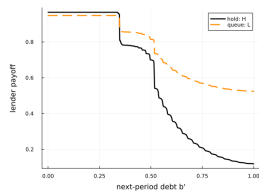
Equilibrium private borrowing policy

Debt policy, low growth



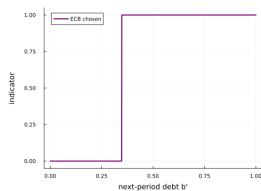
Baseline Policy: ECB Facility and Default

ECB facility payoff, low growth and bad sunspot

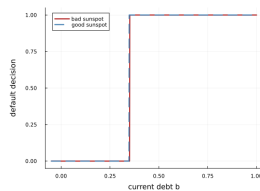


Hold-versus-facility payoff comparison

Schedule points supported by ECB

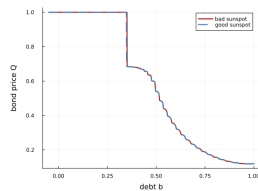


Default policy, low growth

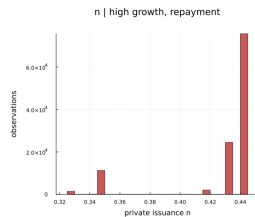
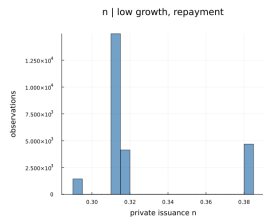


Default policy and bond price

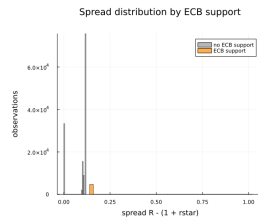
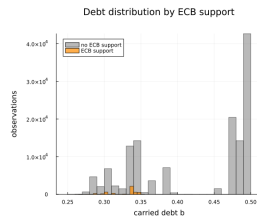
Bond price, low growth



Baseline Simulation: Issuance and ECB Support

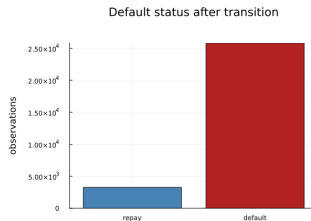
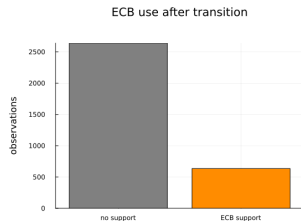
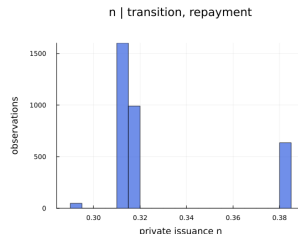
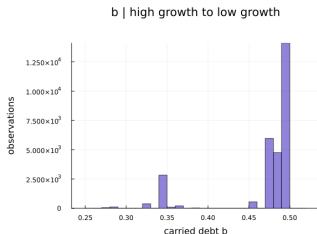


Simulated distribution of private issuance



Outcomes with and without ECB support

Baseline Simulation: Growth-Transition Outcomes



Simulated outcomes around $g_H \rightarrow g_L$ transitions

Baseline Table

Table: ECB baseline simulated moments

Moment	$\pi = 50\%$, $\sigma^\ell = 4\%$
<i>First moments</i>	
avg(spread)	0.0879
avg(qb/y)	0.3996
avg(f/y)	0.4507
avg(n/y)	0.3996
avg(n_l/y)	0.0000
avg(l/y)	0.0000
avg(b/y)	0.4080
avg(tb/y)	0.0084
default rate	0.1489
<i>Low-growth state</i>	
avg(spread)	0.0290
avg(qb/y)	0.3382
avg(f/y)	0.3616
avg(n/y)	0.3382
avg(n_l/y)	0.0000
avg(l/y)	0.0000
avg(b/y)	0.3352
avg(tb/y)	-0.0030
default rate	0.3886
<i>High-growth state</i>	
avg(spread)	0.1008
avg(qb/y)	0.4130
avg(f/y)	0.4702
avg(n/y)	0.4130
avg(n_l/y)	0.0000
avg(l/y)	0.0000
avg(b/y)	0.4240
avg(tb/y)	0.0109
default rate	0.0000

Policy Table

Table: ECB policy-grid simulated moments

Moment	$\pi = 0\%$ $\sigma^\ell = 4\%$	$\pi = 25\%$ $\sigma^\ell = 4\%$	$\pi = 25\%$ $\sigma^\ell = 6\%$	$\pi = 25\%$ $\sigma^\ell = 8\%$	$\pi = 25\%$ $\sigma^\ell = 10\%$	$\pi = 50\%$ $\sigma^\ell = 4\%$	$\pi = 50\%$ $\sigma^\ell = 6\%$	$\pi = 50\%$ $\sigma^\ell = 8\%$	$\pi = 50\%$ $\sigma^\ell = 10\%$	$\pi = 75\%$ $\sigma^\ell = 4\%$	$\pi = 75\%$ $\sigma^\ell = 6\%$	$\pi = 75\%$ $\sigma^\ell = 8\%$	$\pi = 75\%$ $\sigma^\ell = 10\%$	$\pi = 100\%$ $\sigma^\ell = 4\%$	$\pi = 100\%$ $\sigma^\ell = 6\%$	$\pi = 100\%$ $\sigma^\ell = 8\%$	$\pi = 100\%$ $\sigma^\ell = 10\%$
<i>First moments</i>																	
avg(spread)	0.0927	0.0948	0.0936	0.0939	0.0934	0.0891	0.0928	0.0958	0.0964	0.0121	0.0204	0.0420	0.0751	0.0052	0.0072	0.0095	0.0100
avg(qb/y)	0.4052	0.4066	0.4058	0.4060	0.4056	0.4010	0.4037	0.4067	0.4072	0.3485	0.3474	0.3626	0.3876	0.3567	0.3542	0.3528	0.3510
avg(f/y)	0.4593	0.4613	0.4600	0.4603	0.4597	0.4528	0.4571	0.4616	0.4624	0.3657	0.3679	0.3926	0.4327	0.3714	0.3697	0.3692	0.3680
avg(n/y)	0.4052	0.4066	0.4058	0.4060	0.4056	0.4010	0.4037	0.4067	0.4072	0.3485	0.3474	0.3626	0.3876	0.3567	0.3542	0.3528	0.3510
avg(n.l/y)	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
avg(l/y)	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
avg(b/y)	0.4148	0.4162	0.4154	0.4155	0.4152	0.4098	0.4127	0.4160	0.4167	0.3529	0.3529	0.3690	0.3958	0.3586	0.3569	0.3564	0.3550
avg(tb/y)	0.0096	0.0097	0.0096	0.0096	0.0096	0.0088	0.0090	0.0094	0.0095	0.0044	0.0054	0.0064	0.0082	0.0019	0.0027	0.0036	0.0030
default rate	0.1491	0.1531	0.1517	0.1521	0.1511	0.1492	0.1527	0.1557	0.1556	0.0521	0.0616	0.0920	0.1334	0.0544	0.0539	0.0536	0.0530
<i>Low-growth state</i>																	
avg(spread)	0.0005	0.0144	0.0098	0.0097	0.0074	0.0286	0.0277	0.0291	0.0280	0.0267	0.0298	0.0335	0.0328	0.0107	0.0159	0.0210	0.0250
avg(qb/y)	0.2861	0.3078	0.3053	0.3032	0.2999	0.3381	0.3329	0.3286	0.3238	0.3611	0.3546	0.3543	0.3480	0.3769	0.3723	0.3693	0.3660
avg(f/y)	0.2963	0.3240	0.3197	0.3174	0.3132	0.3613	0.3555	0.3516	0.3461	0.3848	0.3793	0.3805	0.3736	0.3949	0.3924	0.3913	0.3900
avg(n/y)	0.2861	0.3078	0.3053	0.3032	0.2999	0.3381	0.3329	0.3286	0.3238	0.3611	0.3546	0.3543	0.3480	0.3769	0.3723	0.3693	0.3660
avg(n.l/y)	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
avg(l/y)	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
avg(b/y)	0.3098	0.3204	0.3202	0.3188	0.3173	0.3351	0.3308	0.3274	0.3252	0.3651	0.3589	0.3553	0.3447	0.3723	0.3702	0.3690	0.3670
avg(tb/y)	0.0237	0.0126	0.0149	0.0156	0.0173	-0.0030	-0.0021	-0.0012	0.0014	0.0041	0.0043	0.0010	-0.0033	-0.0046	-0.0021	-0.0002	0.0000
default rate	0.3968	0.4061	0.4020	0.4029	0.3999	0.3954	0.4065	0.4149	0.4146	0.1362	0.1611	0.2457	0.3510	0.1383	0.1379	0.1393	0.1340
<i>High-growth state</i>																	
avg(spread)	0.1112	0.1106	0.1105	0.1107	0.1108	0.1017	0.1056	0.1084	0.1093	0.0050	0.0160	0.0450	0.0860	0.0025	0.0029	0.0038	0.0030
avg(qb/y)	0.4291	0.4261	0.4260	0.4265	0.4270	0.4142	0.4177	0.4214	0.4229	0.3424	0.3441	0.3656	0.3979	0.3468	0.3453	0.3448	0.3430
avg(f/y)	0.4920	0.4884	0.4883	0.4889	0.4894	0.4719	0.4771	0.4823	0.4843	0.3563	0.3626	0.3970	0.4479	0.3599	0.3585	0.3584	0.3570
avg(n/y)	0.4291	0.4261	0.4260	0.4265	0.4270	0.4142	0.4177	0.4214	0.4229	0.3424	0.3441	0.3656	0.3979	0.3468	0.3453	0.3448	0.3430
avg(n.l/y)	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
avg(l/y)	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
avg(b/y)	0.4358	0.4352	0.4346	0.4349	0.4350	0.4254	0.4288	0.4328	0.4340	0.3470	0.3501	0.3740	0.4090	0.3519	0.3504	0.3502	0.3490
avg(tb/y)	0.0067	0.0091	0.0086	0.0084	0.0080	0.0112	0.0112	0.0114	0.0111	0.0046	0.0059	0.0084	0.0112	0.0050	0.0051	0.0054	0.0050
default rate	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0030	0.0023	0.0008	0.0003

Appendix: Reported Variables

$$f_t = g_t b'_t, \quad q_t = \frac{1}{R_t}, \quad n_t = q_t f_t = \frac{g_t b'_t}{R_t}, \quad q_t b_t = \frac{b_t}{R_t}.$$

$$P_t = \frac{1}{1 + r^* + \sigma^\ell} \quad \text{only when} \quad d_t = 0, \quad g_t = g_L, \quad s_t = s_b.$$

- f_t is face value of newly issued private debt.
- n_t is private proceeds raised today.
- ECB intervention changes the private price schedule; it does not create direct sovereign LOLR debt.
- Therefore, the ECB table entries for $\text{avg}(n_\ell/y)$ and $\text{avg}(\ell/y)$ are zero.